

5. [Maximum mark: 7]

A particle P moves in a straight line. The velocity $v \text{ m s}^{-1}$ of P, at time t seconds is given by $v(t) = e^{-\sin t} \cos(2t)$, for $0 \leq t \leq 5$.

- (a) Find the maximum speed of P. [2]
- (b) Find the total distance travelled by P. [2]
- (c) Find the acceleration when P changes direction for the **second** time. [3]

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